Lec. 1

التسلسل العام:

1) Definitions:
Drug → Medication → pharmacy → NCE

2) Drug Bascis: Naming → Classification → by resource / by dispensing

3) Pharmacy practice:
Drug services → Routs of administration → scope

4) Development of pharmacy profession
History → Advances → challenges

5) patient & professional Roles
Direct VS Indirect patient services → Ten-Star pharmacist

1) Definitions:

Drug → Medication → pharmacy → NCE

Drug: active substance examples:- Acetyl Salicylic acid
- Paracetamol

Medication: the whole dosage form examples: - Tylenol
- Losec

New chemical Entity (NCE): a drug contains an active moiety that has still never been approved by the FDA.

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2) Drug Basics: Naming → Classification → by resource / by dispensing

Naming	Definitions
• Chemical Name	Chemical structure
• Generic Name	Given during early investigation
• official Name	Given in pharmacopoeia
• Brand Name	Trade name in Market

classification of Mediciations according to dispensing:

- prescribed: Principal method of dispensing

Physician: Diagnose + prescribe & pharmacist: Dispense according to prescription.

- Non-prescribed: pharmacist can dispense for minor alignments

Red Flag: ⚠
- Blood in sputum
- Chang in color/odor of "faces"

→ اذا هذي العلامات موجودة → Pharmacist must direct patient to physician

classification of Drugs according to resources: تعريفا مهم

1) Natural
Drugs obtained from natural Resources (galenical products)

2) Semisynthetic
originally from natural resources with some modifications in their chemical structure
Natural origin + chemical modification

3) Synthetic
Totally synthesized chemically

3) Pharmacy practice:

Drug services → Routs of administration → scope

Drug services in the pharmacy profession:

- drug safety
- quantitative drug analysis
- drug distribution
- quality control

Routs of Drugs:

Routs	Advantages	Disadvantages	Exampels
Oral	Safe, cheap, convenient	First-pass metabolism	—
Parentral	Immediate effect (suitable for emergencies)	Risk of infection	—
Into the skin Routes: Subcutaneous / Intradermal	self-admin + long action	Variable drug absorption dependent of blood flow	—
Inhalation / Pulmonary	Rapid + Reaches site	may stimulate the cough reflex	—
Rectal	used for patient unable to swallow, vomiting or unconscious	• uncomfortable and messy to use • Inconvenient for the patient	
Transdermal	• convenient + long action	Local irritation	—
Sublingual	Rapid + avoids first pass	Few drugs + taste/irritation	• Tablets: glyceryl trinitrate SL tablets. • Films: suboxone SL films • sprays: glyceryl trinitrate SL spray

Scope of pharmacy profession:

Phamacist = Select / Labeling / dispence
Monitor / Information

But pharmacy profession does not inlude: Dignose / administer / Lab analyze

4) Development of pharmacy profession

History → Advances → challenges

History:

Era	Main Foucs:
Traditional	traditional profession was pharmacognacy → galenical pharmacy (Nutural drugs ؛ ؛)
Scentific	Pharmacology → studing drugs and their mechanism of action.
Clinical	Combination → traditional roles and new roles
pharmaceutical-care	patient-centerted care

↳ The Goal: ↑ Health-related quality of life + positive clinical outcomes + economic expenditures

Advances:

Traditional role focus on → Drugs /products

↓

Modern role focus on → patients instead of products

↓

Future phamacists will face → more → challenges → responsibilities + opportunities

Challenges:

- Increased chronic diseases
- Control healthcare cost
- quality of life

5) Patient & professional Roles

Direct VS Indirect patient services → Ten-Star pharmacist

Direct: Personal contact with patient	Indirect: not directly related to dignose/treatment
• Immunization	• Family planning
• Preventive care	• Billing
• Wellness screening	• Reporting
• MTM	• National screening
• Chronic disease management	• IT
• Point_care testing	• Care team coordination
• Signposting	• General administration
• Home care/delivery	
• Counseling	

Ten-Star Pharmacist:

Care-giver	Caring services
Decision-maker	Appropriate + cost-effective resources
Communicator	Between physician and patient
Leader	Multidisciplinary caring situations
Manager	Manage resources + information
Life-long learner	Learning throughout career
Teacher	Educate future Pharmacist
Entrepreneur	manage a new pharmacy business (investment من عندي عكس يعني)
Researcher	Investigation new molecules + design new models of practice
Agent for positive change	Improve patient care + pharmacy services + Collaboration

حكى إلا طلاع عليها !

ترجمة

(NCE): a drug contains an active moiety that has still never been appoved by the FDA

Pharmacy Orientation – Chapter 1: Drug and Pharmacy Profession (Summary)

Key Definitions

Drug	A chemical substance that can produce a biological effect in the body.
Medication	A drug product that is prepared and marketed for use (has a trade name, dosage form, strength). ← Example: Tylenol – Losec
Pharmacy	The profession concerned with the preparation, dispensing and appropriate use of medications.

Naming of Drugs (Example: Paracetamol)

Chemical name	N-acetyl-para-aminophenol
Official name (USP)	Acetaminophen
Generic name (INN)	Paracetamol
Brand / Trade names	Tylenol®, Panadol®, Abimol®, Cetal®

New Chemical Entity (NCE) Process

تعريف فقط

Discovery
- Identify and
- synthesize a new compound.

Preclinical
- In vitro & animal studies.
- Evaluate efficacy, safety and toxicity.

IND
- Submit Investigational New Drug application to FDA.

Clinical Trials
- Phase I
- Phase II
- Phase III

NDA
- Submit New Drug Application to FDA (all data from preclinical and clinical).

FDA Approval
- Review safety, efficacy and manufacturing quality.

Postmarketing
- Drug is marketed and used in the general population.
- Monitor long-term safety and effectiveness.

Survey collecting
- Collect data from real-world use (e.g. adverse effects, effectiveness).
- Continue pharmacovigilance.

Classification of Drugs According to the Resource

1. Natural Drugs (plant, animal, directly)
Obtained directly from natural sources (plants, animals, microorganisms or minerals).

Examples:
- Morphine (plant)
- Codeine (plant)
- Digoxin (plant)
- Caffeine (plant)
- Nicotine (plant)
- Snake venom (animal)
- Selenium (mineral)

2. Semisynthetic Drugs
Chemically modified natural compounds derived from natural sources.

Examples:
- Amoxicillin (derived from penicillin)
- Heroin (derived from morphine)

3. Synthetic Drugs
Completely synthesized in the laboratory (not derived from natural sources).

Examples:
- Diclofenac (anti-inflammatory)
- Omeprazole (anti-ulcer)
- Fusidic acid (antibiotic)
- Benzodiazepines

Examples of Drugs and Their Uses

Drug	Use / Classification
Morphine	Natural (analgesic)
Codeine	Natural (analgesic)
Digoxin	Natural (cardiac)
Caffeine	Natural (stimulant)
Nicotine	Natural (stimulant)
Snake venom	Natural (animal source)
Selenium	Natural (mineral)
Amoxicillin	Semisynthetic (antibiotic)
Heroin	Semisynthetic (analgesic)
Diclofenac	Synthetic (anti-inflammatory)
Omeprazole	Synthetic (anti-ulcer)
Fusidic acid	Synthetic (antibiotic)
Benzodiazepines	Synthetic (anxiolytic/sedative)

Dosage Forms and Routes of Administration (Examples)

Route	Examples from the Lecture	Key Point
Sublingual	Glyceryl trinitrate (tablet, spray) Suboxone (film)	Rapid absorption, bypasses first-pass metabolism.
Transdermal	Flupentixol (patch)	Long duration of action.
Rectal	(e.g. suppositories, enemas)	Alternative route.
Inhalation	(not specific example in slides)	Rapid effect.
Oral	(most common)	May be affected by first-pass metabolism.

Drug vs Medication (Comparison)

Feature	Drug = active substance	Medication = dosage form
Meaning	Chemical substance with biological effect	Prepared product for use a specific dosage form
Form	Pure chemical substance	Finished dosage form (tablet, capsule, etc.)
Name	Chemical / generic name	Brand (trade) name
Example (paracetamol)	Paracetamol	Tylenol®, Panadol®, Abimol®, Cetal®

Types of Names (Comparison)

Type	Definition	Example
Chemical name	Describes chemical structure	[illegible]
Official name (USP)	name given to drug in the pharmacopoeia	
Generic name (INN)	Non-proprietary name	Paracetamol
Brand / Trade name	Marketed name by manufacturer	Tylenol®, Panadol®, Abimol®, Cetal®

Important Points

- NCE: from discovery → preclinical → IND → clinical trials → NDA → FDA approval → postmarketing → survey collecting (pharmacovigilance).
- Know the definitions: drug, medication, pharmacy.
- Be able to identify the type of name (chemical, official, generic, brand).
- **Classify drugs according to their resource with correct examples.**
- Know the therapeutic use of the example drugs (diclofenac = anti-inflammatory, omeprazole = anti-ulcer, fusidic acid = antibiotic, etc.).
- Know the examples of dosage forms and routes (glyceryl trinitrate, suboxone, flupentixol).
- Understand the difference between drug and medication with examples.

Pharmacy Orientation – Lecture 1: Drug and Pharmacy Profession (Part 2)

Routes of Drug Administration

Oral



Description
- Most frequently used route.

Examples
- Tablets (immediate-release, enteric-coated, modified-release)
- Capsules
- Granules
- Powders
- Syrups, elixirs
- Suspensions

(solid dosage / liquid dosage)

Advantages
- Cheap
- Generally safe
- Simple and convenient
- Self-administered
- Non-invasive

Disadvantages
- Variable absorption
- Affected by GI motility, gastric emptying and food
- First-pass metabolism

Parenteral / Intravenous (IV)



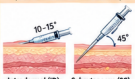
Description
- Administration into tissues by injection.
- IV: into a vein.

Examples
- Injection
- Emulsion injection
- Solution for injection
- Solution for infusion

Advantages
- Immediate effect
- Suitable for emergencies and unconscious patients
- Avoids first-pass metabolism
- Precise and predictable plasma levels

Disadvantages
- Possible anaphylaxis
- Infection risk
- Painful

Into the Skin (Subcutaneous or Intradermal)



Advantages
- Self-administered → skin
- Some drugs have long duration of action (e.g. flupentixol)
- Low risk of systemic infection

Disadvantages
- Variable absorption depending on blood flow
- Only small volume can be given

Pulmonary / Inhalation



Description
- Delivered to the lung for local or systemic effect.

Examples
- Metered-dose inhalers (MDIs)
- Dry powder inhalers

Advantages
- Rapid onset
- Reaches site of action

Disadvantages
- Proper technique required
- Only a small number of drugs can be given this way
- May stimulate cough reflex

Rectal



Description
- For localized or systemic effects.

Examples
- Suppositories
- Enemas

Advantages
- Self-administered → skin
- Local effect possible
- Useful if patient cannot swallow / is vomiting / unconscious
- Reduced first-pass metabolism

Disadvantages
- Uncomfortable and messy
- Inconvenient
- Slow and erratic absorption
- Poorly accepted

Transdermal



Description
- Absorption through the skin.

Examples
- Transdermal patches
- Transdermal gels

Advantages
- Convenient
- Long duration of action
- Less frequent application (e.g. 24 h / 72 h)
- Self-administered
- Avoids first-pass metabolism
- Steady plasma concentration
- Slow absorption

Disadvantages
- Can be expensive
- Local irritation

Sublingual



Description
- Dosage form under the tongue. Absorbed directly to systemic circulation.

Examples (SL = sublingual)
- Glyceryl trinitrate (SL tablet)
- Suboxone (SL film)
- Glyceryl trinitrate (SL spray)

Advantages
- Rapid absorption
- Quick onset
- Avoids first-pass metabolism
- Self-administered
- Convenient
- Helpful for people who have difficulty swallowing tablets

Disadvantages
- Most drugs are not available as sublingual forms
- Taste may be disliked
- Keeping the dosage form under the tongue may be inconvenient
- Oral mucosal irritation

Examples of Areas Included in the Pharmacy Profession

- Community Pharmacy
- Hospital Pharmacy
- Clinical Pharmacy
- Compounding Pharmacy
- Academic Pharmacy
- Consultant Pharmacy
- Internet Pharmacy
- Pharmacoeconomics
- Pharmacy Administration
- Industrial Pharmacy
- Health Center Pharmacy
- Long-term Care Pharmacy
- Drug Information Center
- Government Pharmacy
- Office-based Management
- Regulatory Affairs
- Pharmacoepidemiology
- Pharmacy Benefit Management

Scope of Pharmacy Profession

INCLUDES
- Interpretation of prescription orders
- Compounding
- Labeling
- Dispensing of drugs and devices
- Drug product selection and drug utilization reviews
- Patient monitoring and interventions
- Provision of information related to medications and devices

DOES NOT include
- Diagnosis
- Drug administration
- Laboratory analysis

History of the Pharmacy Profession

Traditional Era
- Pharmacognosy
- Medicinal function of natural products from animal, plant, and mineral sources
- Dominated by formulation and dispensing of drugs from natural sources.

Scientific Era
- Drug development, scientific testing, mass production
- Backbone = pharmacology (scientific study of drugs and their mechanism of action).

Clinical Era (1960s)
- Combination of traditional roles + dispenser of drug information.

Pharmaceutical-care Era
- Patient-centered, outcomes-oriented practice.
- Pharmacist works with patient and other healthcare providers to promote health, prevent disease, and assess / monitor / initiate / modify medication use to assure safe and effective therapy.
- Goal = optimize health-related quality of life and achieve positive clinical outcomes within economic expenditures.

Advances in the Pharmacy Profession

Traditional role
Manufacture and supply medicines.
Focus on DRUG / PRODUCT

Increasing health demands shift focus to PATIENTS.

Now
Focus on PATIENTS

- Expanding responsibilities for healthcare services.
- Increasing opportunities in the pharmacy profession.

Challenges Facing the Pharmacy Profession

- Increased incidence of chronic diseases.
- Increased complexity, number, and sophistication of medications and related products/devices.
- Increased emphasis on primary and preventive health services.
- Home health care and long-term care.
- Improving access to health care.
- Controlling health care cost.
- Assuring patients' quality of life.

Patient Services Included in the Pharmacy Profession

Direct Patient Services
Interventions with personal contact with patients.

Examples:
- Immunizations
- Preventative care services
- Wellness screenings
- Collaborative care services
- Medication therapy management (MTM)
- Chronic disease management
- Point-care testing
- Signposting
- Home-based caring
- Home-delivery
- Counseling

Indirect Patient Services
Health care services not directly related to diagnosis or treatment of a patient.

Examples:
- Family planning program
- Billing
- Reporting
- National screening
- Information technology
- Care team coordination
- General administration

Ten-Star Pharmacist

1	Care-giver	Provides caring services. Appropriate, efficacious, and cost-effective use of resources.
2	Decision-maker	
3	Communicator	Ideal position between physician and patient.
4	Leader	Functions in multidisciplinary caring situations.
5	Manager	Effectively manages resources and information.
6	Life-long learner	Commitment to continuous learning throughout career; Assists with education and training of future pharmacists.
7	Teacher	
8	Entrepreneur	Can make market survey, plan, direct, and manage a new pharmacy business.
9	Researcher	Investigates new molecules and designs new models of practice.
10	Agent for positive change	Leads change to improve patient care, pharmacy services, and interprofessional collaboration.